

DIGITAL BANKING (E-Banking)

1. Introduction to Digital Banking

Digital Banking refers to the delivery of banking services through digital platforms such as internet banking, mobile banking applications, ATMs, payment cards, UPI, digital wallets and other electronic channels. It reduces dependence on traditional paper-based banking methods such as cheques, pay-in slips and demand drafts. Digital banking allows customers to perform banking activities such as fund transfer, bill payment, account management, investment and financial transactions anytime and anywhere through electronic devices.

The Payment and Settlement Systems Act, 2007 defines electronic fund transfer as a transfer of funds initiated through electronic instructions to debit or credit an account maintained with a bank. It includes transactions through point-of-sale terminals, ATMs, internet banking, telephone banking and card-based payments.

Digital Banking has become a major component of the Digital India programme of the Government of India, which aims to create a digitally empowered society and knowledge economy. The vision of Digital India includes a “**Faceless, Paperless and Cashless**” economy.

2. Advantages of Digital Banking

2.1 Anytime and Anywhere Banking

Digital banking provides customers with 24×7 access to banking services. Customers can transfer money, check account balances, download statements and pay bills without visiting a bank branch.

2.2 Faster Fund Transfer

Digital platforms provide multiple options for transferring funds, including NEFT, RTGS, IMPS and UPI. These systems have reduced dependency on physical instruments like cheques and demand drafts.

2.3 Reduction in Transaction Cost

Digital banking reduces operational costs for banks by decreasing paperwork, manual processing and physical infrastructure requirements.

2.4 Transparency and Accountability

Electronic transactions create digital records, improving transparency, tracking and accountability in financial activities.

2.5 Reduction in Cash Dependency

Digital payments reduce the risks associated with carrying large amounts of cash and help in controlling fake currency circulation and unaccounted transactions.

2.6 Financial Inclusion

Digital banking helps extend banking services to rural and underserved populations through technologies such as Micro ATMs, Aadhaar Enabled Payment System (AePS), USSD banking and Business Correspondents.

3. Evolution of Digital Payments in India

The digital payment ecosystem in India has developed gradually through various technological innovations.

Major Milestones:

MICR Clearing (1980s):

Magnetic Ink Character Recognition (MICR) introduced electronic processing of cheques by capturing cheque images and MICR data.

Electronic Clearing Service (ECS) and Electronic Funds Transfer (EFT) – 1990s:

These systems enabled electronic movement of funds between bank accounts.

Debit and Credit Cards – 1990s:

Banks introduced payment cards, providing customers with electronic payment facilities.

National Financial Switch (NFS) – 2003:

NFS enabled interconnectivity among ATMs across India.

RTGS and NEFT – 2004:

These became major electronic fund transfer systems for domestic banking transactions.

Cheque Truncation System (CTS) – 2008:

CTS replaced physical movement of cheques with electronic cheque image processing.

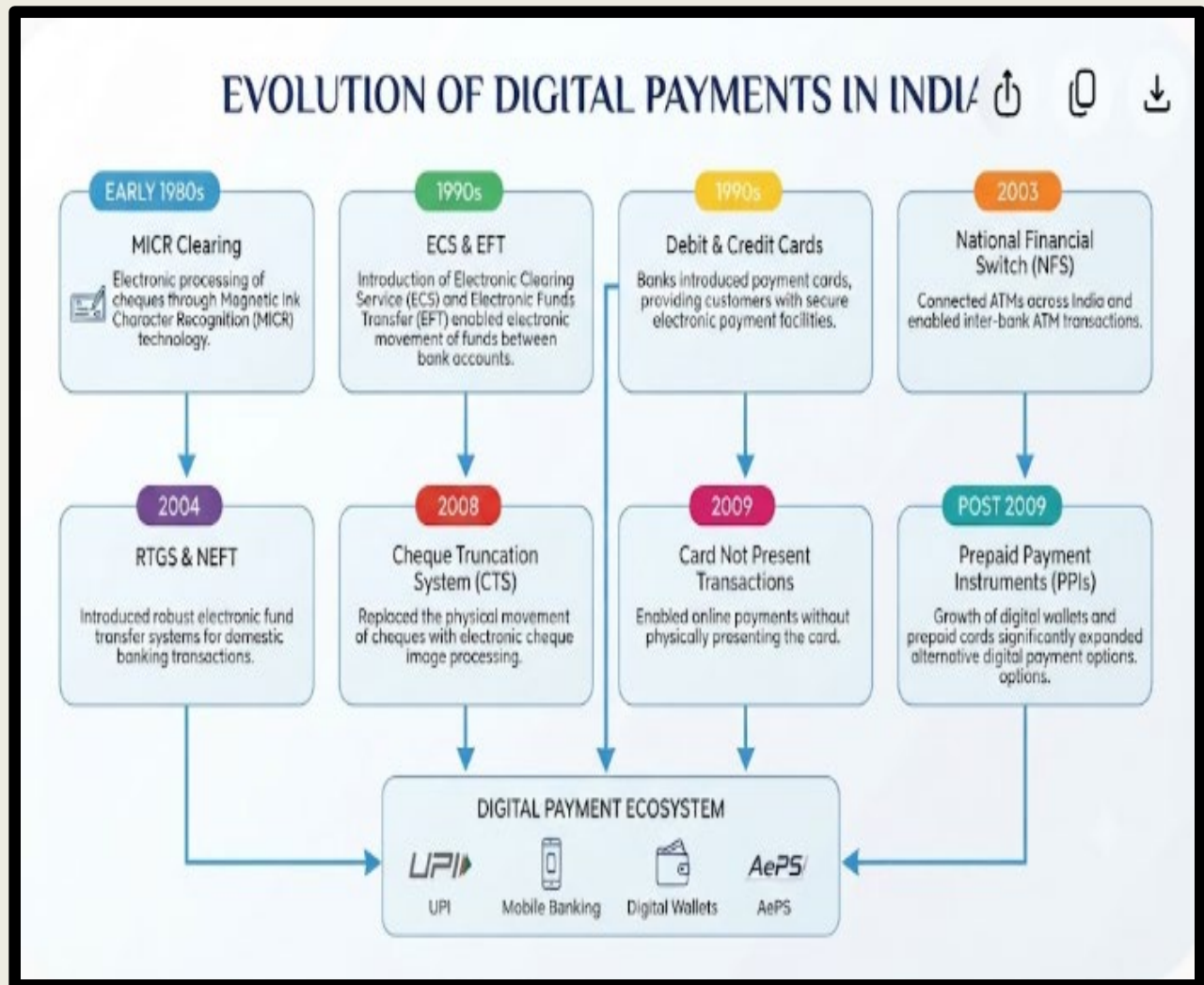
Card Not Present Transactions – 2009:

Enabled online payments without physically presenting the card.

Prepaid Payment Instruments (PPIs):

Digital wallets and prepaid cards expanded digital payment options.

Evolution of Digital Payment Instruments



4. Banking Cards

Banking cards are electronic payment instruments that provide customers with secure, convenient and efficient payment facilities. Major types include:

1. Debit Cards
2. Credit Cards
3. Prepaid Cards

Card payment systems in India include RuPay, Visa and MasterCard. These cards use security mechanisms such as PIN and OTP-based two-factor authentication.

5. Debit Card

A debit card is a payment card directly linked with the customer's bank account. When a transaction is performed, money is automatically deducted from the account balance.

Debit cards eliminate the need to carry physical cash and can be used at:

- ATMs
- Point of Sale (PoS) terminals
- Online shopping websites
- Mobile wallets
- E-commerce platforms

Features of Debit Card

- Linked directly with savings/current account.
- Payment is made using available account balance.
- Requires PIN and OTP authentication.
- Can be used domestically and internationally.
- Usually has daily transaction limits.

6. Credit Card

A credit card is issued by a financial institution that allows customers to borrow money up to a predetermined credit limit. The limit depends on the customer's credit profile and repayment capacity.

Credit cards can be used for:

- Shopping
- Online payments
- ATM cash withdrawal
- Bill payments
- International transactions

Features of Credit Card

- Provides short-term credit facility.

- Payment is made later through monthly billing.
- Interest is charged if outstanding dues are not paid within the due period.
- Helps build credit history.
- Provides reward points, cashback and offers.

7. Difference Between Debit Card and Credit Card

Basis	Debit Card	Credit Card
Source of Money	Customer's bank account	Borrowed money from issuer
Payment Time	Immediate deduction	Payment after billing cycle
Interest	Generally not applicable	Charged on unpaid amount
Credit Score	No direct impact	Affects credit score
Spending Limit	Based on account balance	Based on credit limit

8. Automated Teller Machines (ATM)

An Automated Teller Machine (ATM) is an electronic banking device that allows customers to perform basic banking transactions without visiting a bank branch. ATMs mainly provide services such as cash withdrawal, balance enquiry, mini statements, PIN change and other banking facilities.

ATMs have transformed from simple cash dispensing machines into advanced digital banking terminals capable of providing multiple services, including cardless cash withdrawal and other banking facilities.

Types of ATMs

8.1 Bank-Owned ATMs

These ATMs are owned and operated by the bank itself. They display the logo of the concerned bank and provide services mainly to that bank's customers.

8.2 Brown Label ATMs (BLA)

Brown Label ATMs are operated by third-party service providers. The bank owns the responsibility of branding and cash management, while certain operational activities such as maintenance, electricity and infrastructure are outsourced.

8.3 White Label ATMs (WLA)

White Label ATMs are owned and operated by non-banking companies. These ATMs allow customers of different banks to withdraw cash. They do not display the logo of any particular bank but display the logo of the ATM operator.

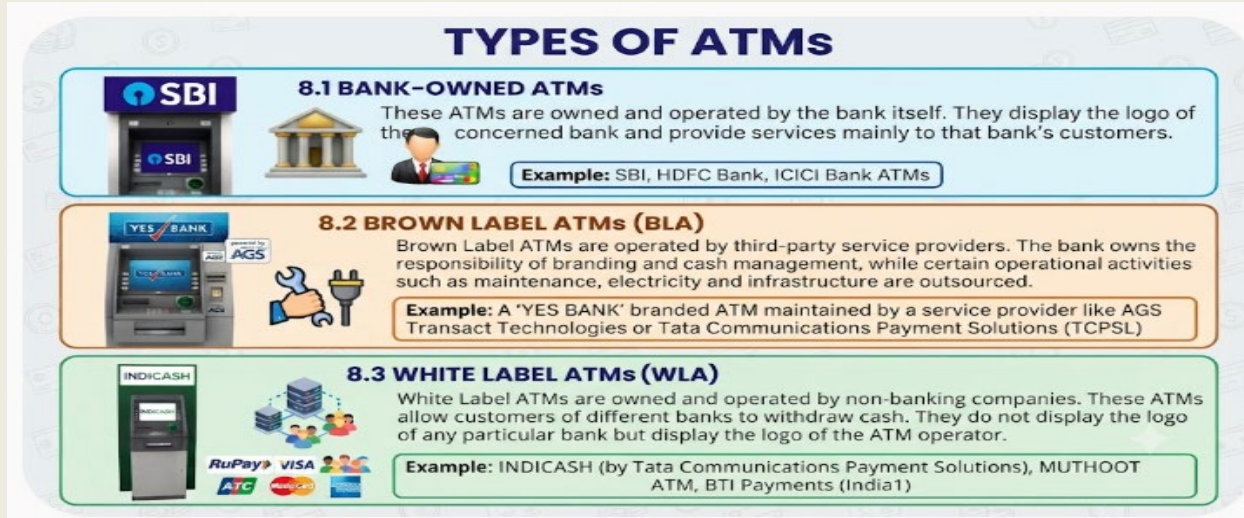
8.4 Micro ATM

Micro ATM is a handheld device used mainly by Business Correspondents (BCs) to provide basic banking services in rural and remote areas.

Services provided:

- Cash deposit
- Cash withdrawal
- Balance enquiry
- Fund transfer

Customers can authenticate transactions using Aadhaar-based biometric verification.



9. National Financial Switch (NFS)

National Financial Switch was launched in 2003 to connect ATMs across India. It enabled customers to access banking services through ATMs of other banks.

NFS was initially developed by IDRBT and later transferred to NPCI in 2009. It provides services such as:

- Cash withdrawal
- Balance enquiry
- PIN change
- Mini statement
- Mobile banking registration
- Aadhaar number seeding

10. National Electronic Funds Transfer (NEFT)

National Electronic Funds Transfer (NEFT) is an electronic fund transfer system that enables individuals, firms, companies and organisations to transfer money electronically from one bank account to another bank account within India. NEFT is a nationwide payment system operated by the Reserve Bank of India (RBI) that facilitates secure and convenient inter-bank fund transfers. It eliminates the need for physical payment instruments such as cheques and demand drafts. NEFT is mainly designed for retail-value transactions, where customers can transfer money for personal, business and institutional purposes.

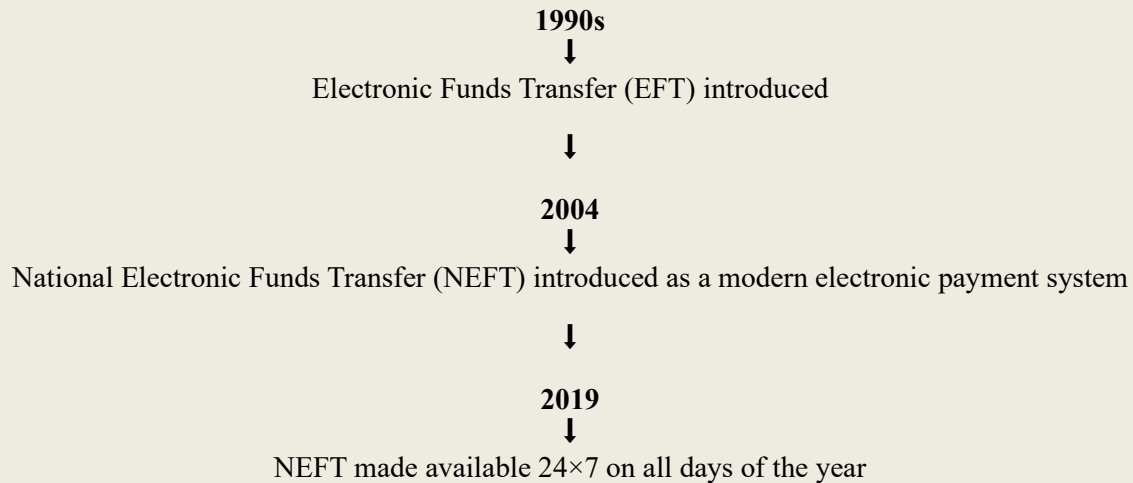
Unlike RTGS, where every transaction is settled individually in real time, NEFT transactions are processed through an electronic clearing mechanism. Earlier NEFT operated through batches, but after technological upgrades, NEFT is available **24×7 throughout the year**, including weekends and bank holidays.

Evolution of NEFT

The concept of electronic fund transfer started in India during the 1990s with the introduction of **Electronic Funds Transfer (EFT)**.

Later, the Reserve Bank of India introduced NEFT as a more advanced and secure electronic payment system.

Timeline:



Objectives of NEFT

The main objectives of NEFT are:

1. Faster Electronic Payments

NEFT provides a convenient method for transferring money electronically without visiting bank branches.

2. Reduce Dependence on Cash and Paper Instruments

It reduces the use of:

- Cash transactions
- Cheques
- Demand drafts

3. Promote Digital Banking

NEFT supports India's movement towards a cashless and digital economy.

4. Secure Interbank Transfer

It provides a safe mechanism for transferring funds between different banks.

Features of NEFT

1. 24×7 Availability

NEFT is available every day of the year, including:

- Sundays
- Bank holidays
- Public holidays

Customers can initiate transactions at any time through internet banking or mobile banking.

2. No Minimum Transfer Limit

There is no minimum amount prescribed by RBI for NEFT transactions.

A customer can transfer even a small amount through NEFT.

Example: A customer can transfer ₹100, ₹500 or ₹1,000 through NEFT depending on bank policies.

3. No Maximum Limit Prescribed by RBI

The Reserve Bank of India has not fixed any maximum transfer limit for NEFT.

However, individual banks may prescribe their own limits based on:

- Risk management policies
- Customer category
- Digital banking rules

4. Interbank Transfer Facility

NEFT allows transfer of money between:

- Same bank accounts
- Different bank accounts

provided both banks participate in the NEFT network.

5. Secure Banking System

NEFT transactions are processed through RBI-controlled payment infrastructure, ensuring secure movement of funds.

6. Suitable for Retail Transactions

NEFT is commonly used for:

- Salary payments
- Loan payments
- Online purchases
- Personal fund transfers
- Business payments

Requirements for NEFT Transaction

To initiate an NEFT transfer, the sender requires:

Sender Details:

- Bank account
- Internet banking/mobile banking facility
- Registered mobile number

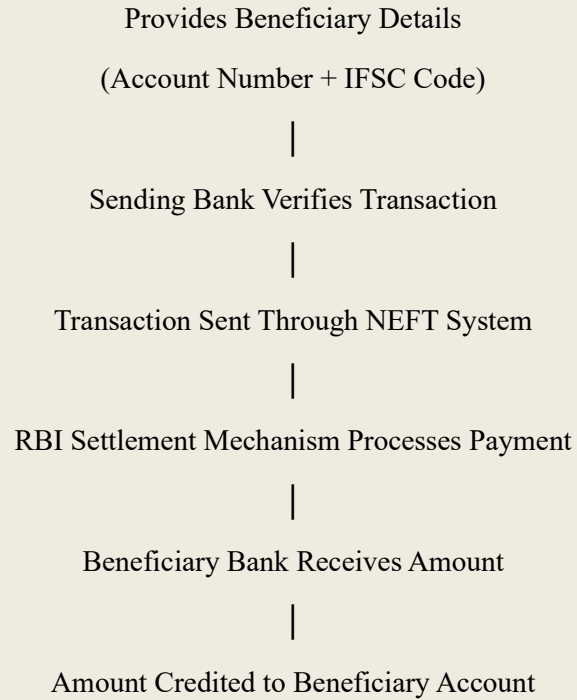
Beneficiary Details:

1. Beneficiary name
2. Beneficiary account number
3. Beneficiary bank name
4. IFSC code of beneficiary branch
5. Transfer amount

NEFT Process Flow

Customer Initiates NEFT Request

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Advantages of NEFT

1. Convenient

Customers can transfer money without visiting bank branches.

2. Cost Effective

NEFT is generally cheaper compared with traditional transfer methods.

3. Safe and Reliable

Transactions are processed through RBI-regulated systems.

4. Paperless Banking

It reduces paperwork and supports digital banking.

5. Nationwide Reach

Money can be transferred to any NEFT-enabled bank branch across India.

Limitations of NEFT

- Requires beneficiary bank details.

- Transaction settlement may take longer compared with instant payment systems like UPI and IMPS.
- Internet/mobile banking facility is required for online transfer.

11. Real Time Gross Settlement (RTGS)

RTGS (Real Time Gross Settlement) is an electronic payment system where fund transfers are processed and settled individually on a real-time basis. The term "Gross Settlement" means that every transaction is settled separately without combining or netting multiple transactions. RTGS is mainly designed for high-value financial transactions where immediate settlement is required.

Meaning of Each Term in RTGS

Real Time-The transaction is processed immediately after receiving payment instructions. There is no waiting period or batch processing.

Gross Settlement

Each transaction is settled individually.

Example:

If Bank A sends three payments:

- ₹5 lakh
- ₹10 lakh
- ₹15 lakh

RTGS settles each transaction separately instead of combining them.

Objectives of RTGS

1. Immediate Settlement

Provides instant transfer of high-value funds.

2. Reduce Settlement Risk

Since transactions are settled immediately, the risk of payment failure is reduced.

3. Improve Banking Efficiency

Provides a reliable mechanism for large financial transactions.

Features of RTGS

1. Real-Time Transfer

Funds are transferred immediately after processing.

2. High-Value Transaction System

RTGS is mainly used for large-value payments.

Examples:

- Corporate payments
- Interbank transfers
- Large business transactions

3. Minimum Transfer Amount

The minimum amount for RTGS transactions is: **₹2 lakh**

4. No Maximum Limit

RBI has not prescribed any maximum transaction limit. Banks may apply internal limits.

5. Operated Through RBI

RTGS transactions are settled through the Reserve Bank of India's payment system.

6. Final and Irrevocable Settlement

Once an RTGS transaction is completed, it cannot normally be reversed.

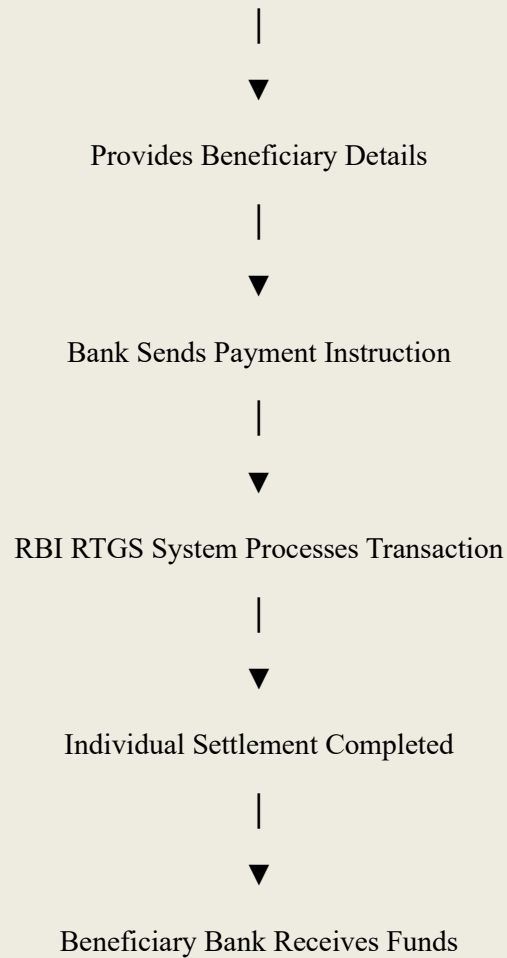
Requirements for RTGS Transaction

Customer requires:

- Bank account
- Internet banking/mobile banking
- Beneficiary details
- IFSC code
- Minimum ₹2 lakh transaction amount

RTGS Process Flow

Customer Initiates RTGS Request



Advantages of RTGS

1. Fast Transfer

Provides immediate transfer of funds.

2. Secure Settlement

Transactions are directly settled through RBI infrastructure.

3. Suitable for Large Payments

Ideal for high-value financial transactions.

4. Reduces Settlement Risk

Immediate settlement reduces credit and liquidity risks.

Difference Between NEFT and RTGS

Basis	NEFT	RTGS
Full Form	National Electronic Funds Transfer	Real Time Gross Settlement
Operated By	RBI	RBI
Nature	Retail payment system	High-value payment system
Settlement Method	Electronic clearing	Individual real-time settlement
Processing	Batch-based electronic processing	Real-time processing
Minimum Amount	No minimum limit	₹2 lakh
Maximum Amount	No RBI maximum limit	No RBI maximum limit
Transaction Speed	Depends on settlement cycle	Immediate
Suitable For	Small and medium transactions	Large-value transactions
Settlement Risk	Comparatively higher	Lower due to immediate settlement

12. IMPS (Immediate Payment Service)

IMPS is an instant interbank electronic fund transfer service developed by NPCI. It allows customers to transfer money immediately through mobile banking, internet banking and ATMs.

Features

- Instant fund transfer.
- Available 24×7.
- Uses account number and IFSC code or MMID.
- Suitable for urgent small-value transfers.

13. Unified Payments Interface (UPI)

Unified Payments Interface (UPI) is an instant payment system developed by NPCI that allows multiple bank accounts to be linked through a single mobile application.

UPI enables:

- Person-to-person transfer
- Merchant payments
- QR code payments
- Collect requests

Features of UPI

- Transfer money without sharing bank account details.
- Uses Virtual Payment Address (VPA).
- Provides two-factor authentication.
- Supports QR-based payments.
- Works through smartphones.

UPI transactions generally have a limit of ₹1 lakh for normal transactions, with higher limits available for certain categories.

UPI Payment Process

1. Download UPI application.
2. Register mobile number linked with bank account.
3. Create Virtual Payment Address.
4. Set UPI PIN.
5. Send or receive money.

14. UPI 123Pay

UPI 123Pay was introduced for feature phone users to enable digital payments without internet connectivity.

It provides four methods:

1. App-based payment
2. Missed call payment
3. IVR-based payment
4. Sound-based proximity payment

15. Mobile Wallets

A mobile wallet is a digital application that stores money electronically and allows users to make payments without using physical cash or cards.

Users can link debit cards, credit cards or bank accounts with wallets.

Examples:

- Paytm
- PhonePe
- Mobikwik
- Amazon Pay

Types of Wallets

Open Wallet

Allows users to make payments, withdraw cash and transfer money.

Semi-Open Wallet

Allows payments but does not permit cash withdrawal.

Closed Wallet

Used only for purchasing goods and services from the issuing company.

16. Internet Banking

Internet banking, also known as online banking or e-banking, is an electronic system that enables customers to perform banking transactions through internet platforms.

Services include:

- Fund transfer
- Balance enquiry
- Account statements
- Bill payments
- Investment services
- Loan applications

17. Bharat Bill Payment System (BBPS)

BBPS is an integrated bill payment platform that allows customers to pay multiple bills through a single system.

Bills covered:

- Electricity
- Water
- Gas
- Telephone
- DTH

It provides instant confirmation after payment.

18. Aadhaar Enabled Payment System (AePS)

AePS is a bank-led digital payment system that allows customers to perform banking transactions using Aadhaar authentication through Micro ATMs or Business Correspondents.

Services:

- Balance enquiry
- Cash withdrawal
- Cash deposit
- Aadhaar-to-Aadhaar transfer
- Bill payments

19. Digital Banking Security Measures

Digital banking requires strong security mechanisms because online transactions face risks such as fraud, phishing and cyber attacks.

Important Security Measures

Two-Factor Authentication

Banks use multiple verification methods:

- Password/PIN
- OTP
- Biometrics

Encryption

Sensitive financial data is protected through encryption technology.

Customer Awareness

Customers should:

- Never share OTP, PIN or passwords.
- Avoid unknown links.
- Use official banking applications.
- Enable transaction alerts.

20. Types of Digital Banking Frauds

Phishing

Fraudulent emails or messages pretending to be from banks to steal customer information.

Vishing

Fraud through phone calls where criminals pretend to be bank officials.

Smishing

Fraud through SMS messages containing malicious links.

Card Skimming

Illegal copying of card information using devices attached to ATMs or payment machines.

Website Spoofing

Creating fake websites similar to genuine banking websites.